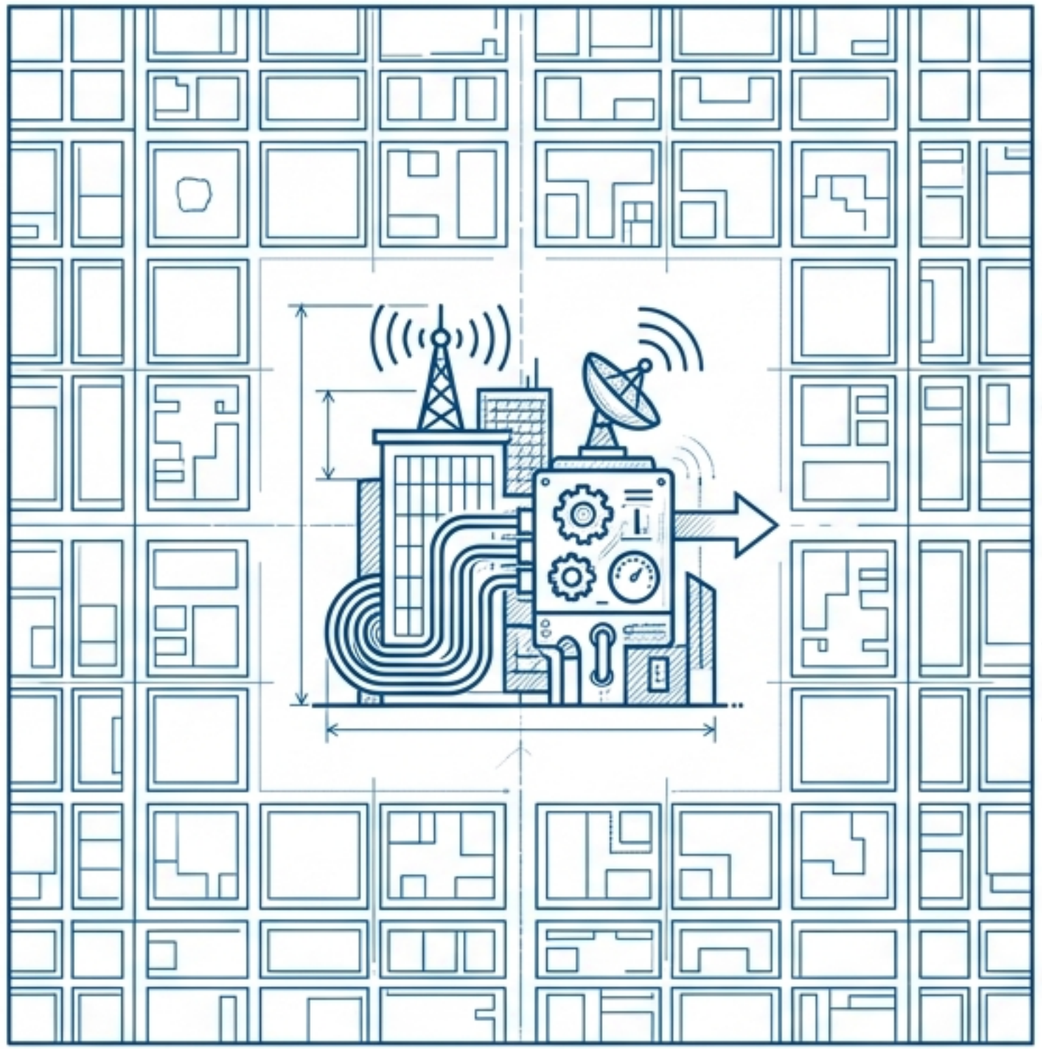


Low-Cost Air PollutionSensor+IoT + Machine Learning

LoRaWAN Connectivity and ANN-Based Calibration

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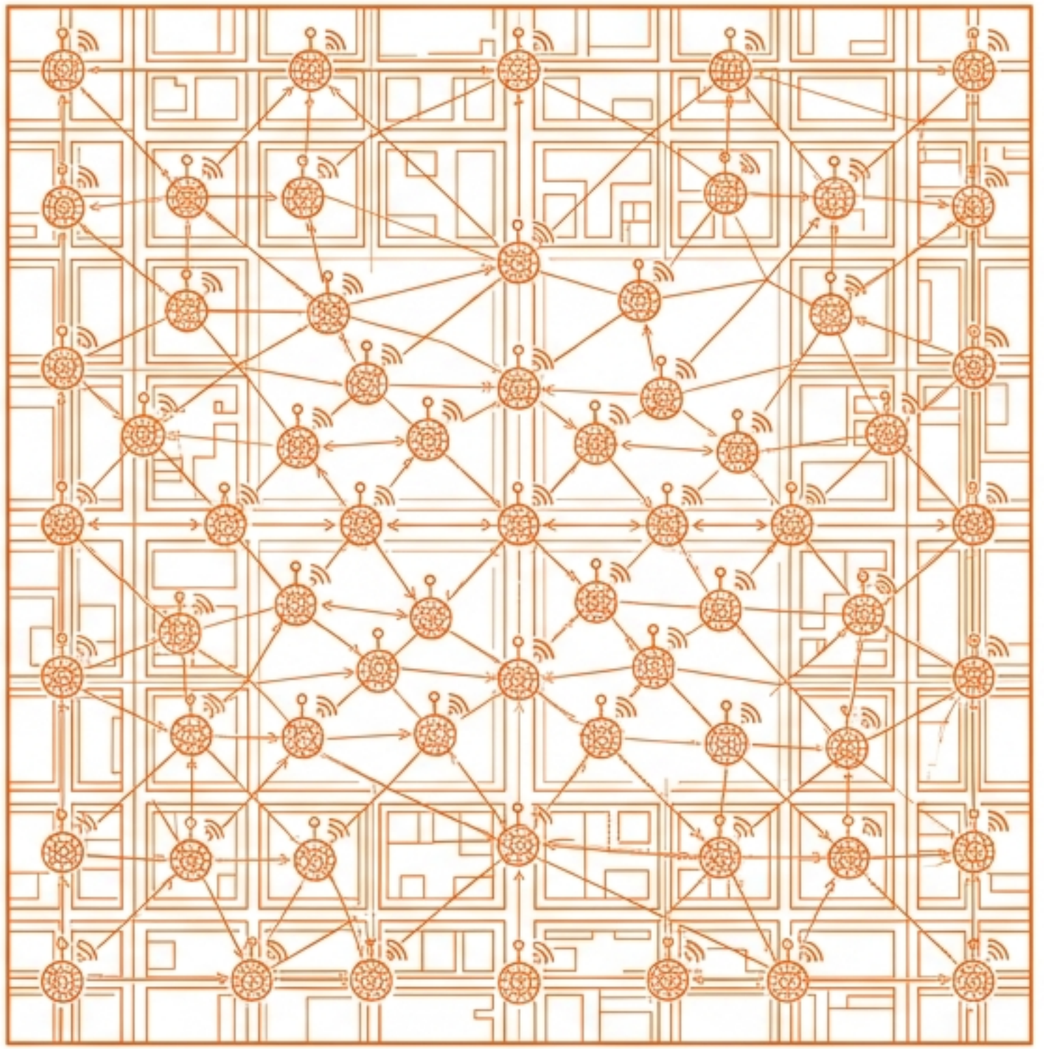
Traditional Monitoring



High Cost → Low Spatial Coverage

EVOLUTION

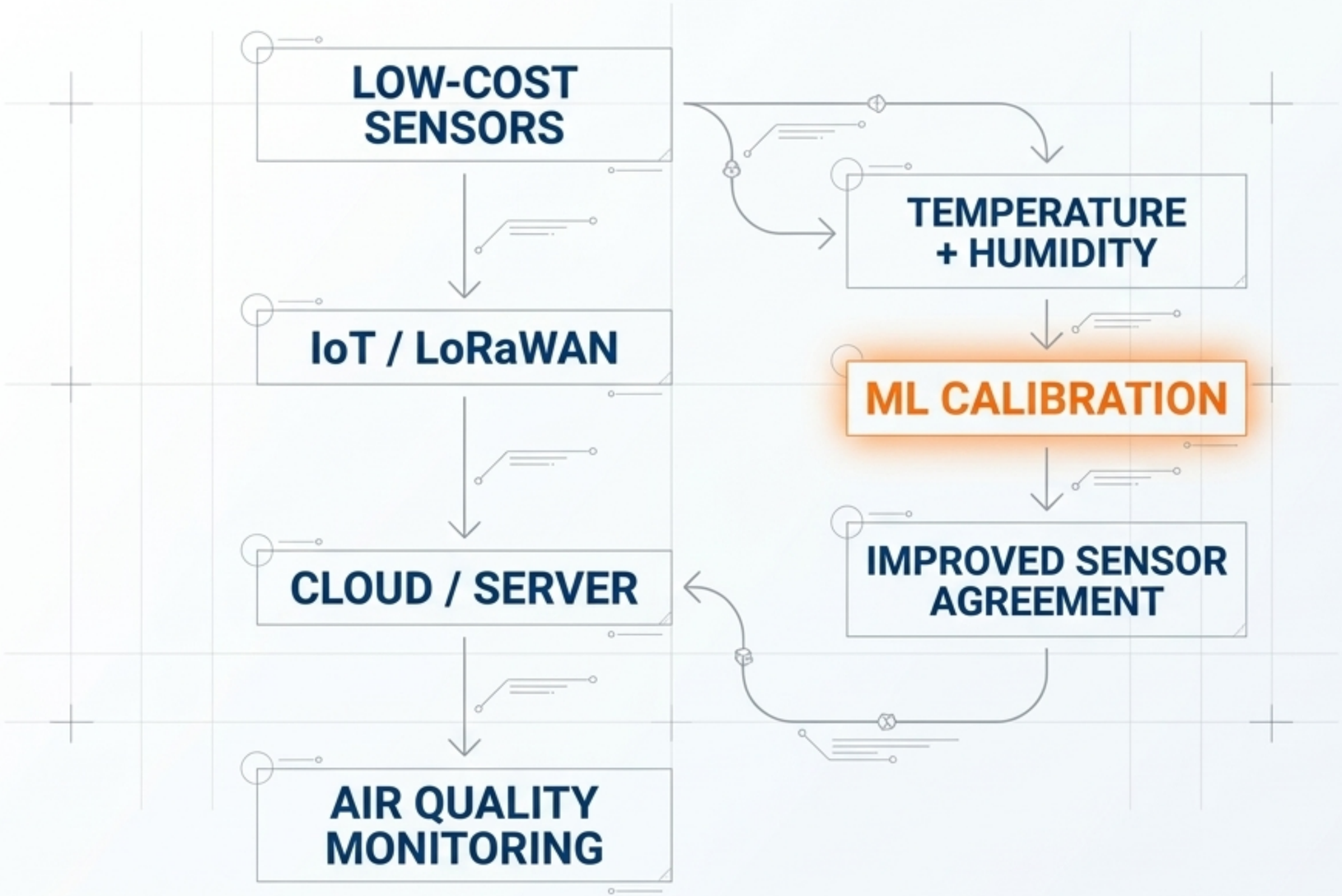
Emerging IoT Paradigm



Low-Cost Network → High-Density Monitoring

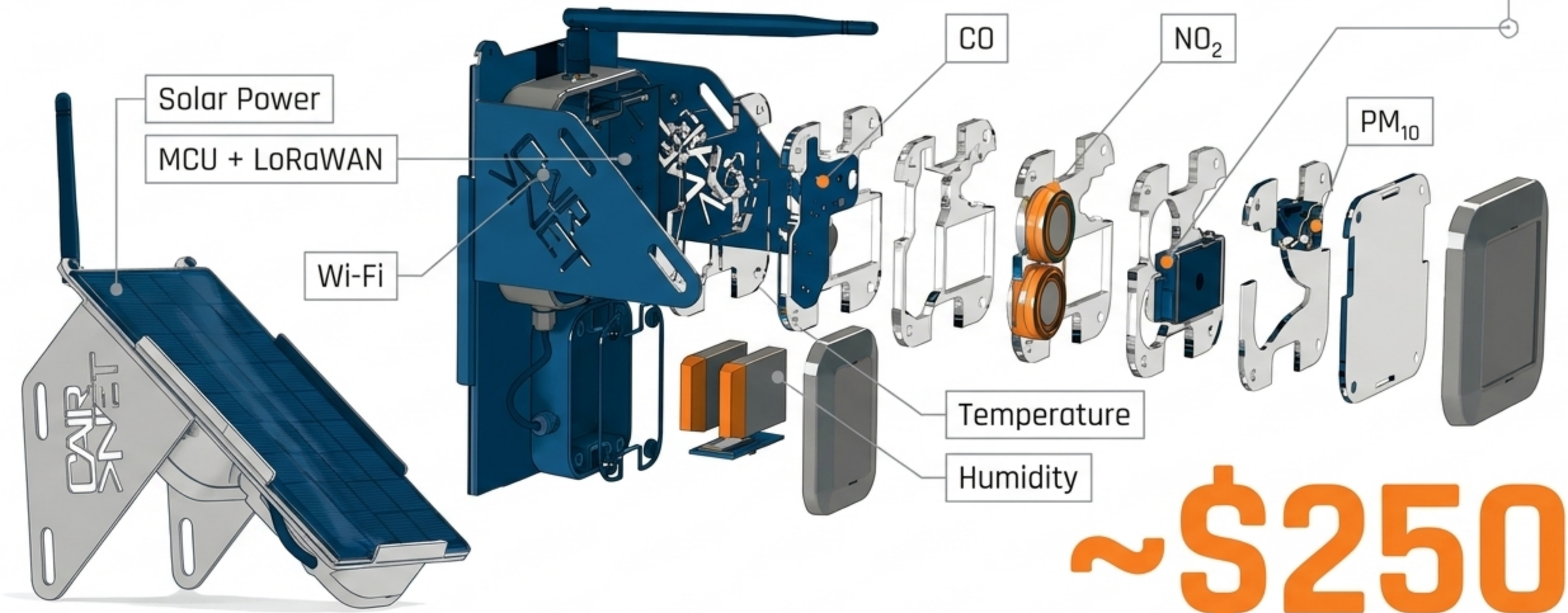
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CONNECTED IoT AIR QUALITY MONITORING BLUEPRINT



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The Physical Edge Layer



~$250

Source: Table I in the paper

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Multi-Pollutant Sensing Strategy



CO & NO_2
→ **Electrochemical Sensors**
(Linear response)



PM_{10}
→ **Optical Sensor**
(Light scattering)

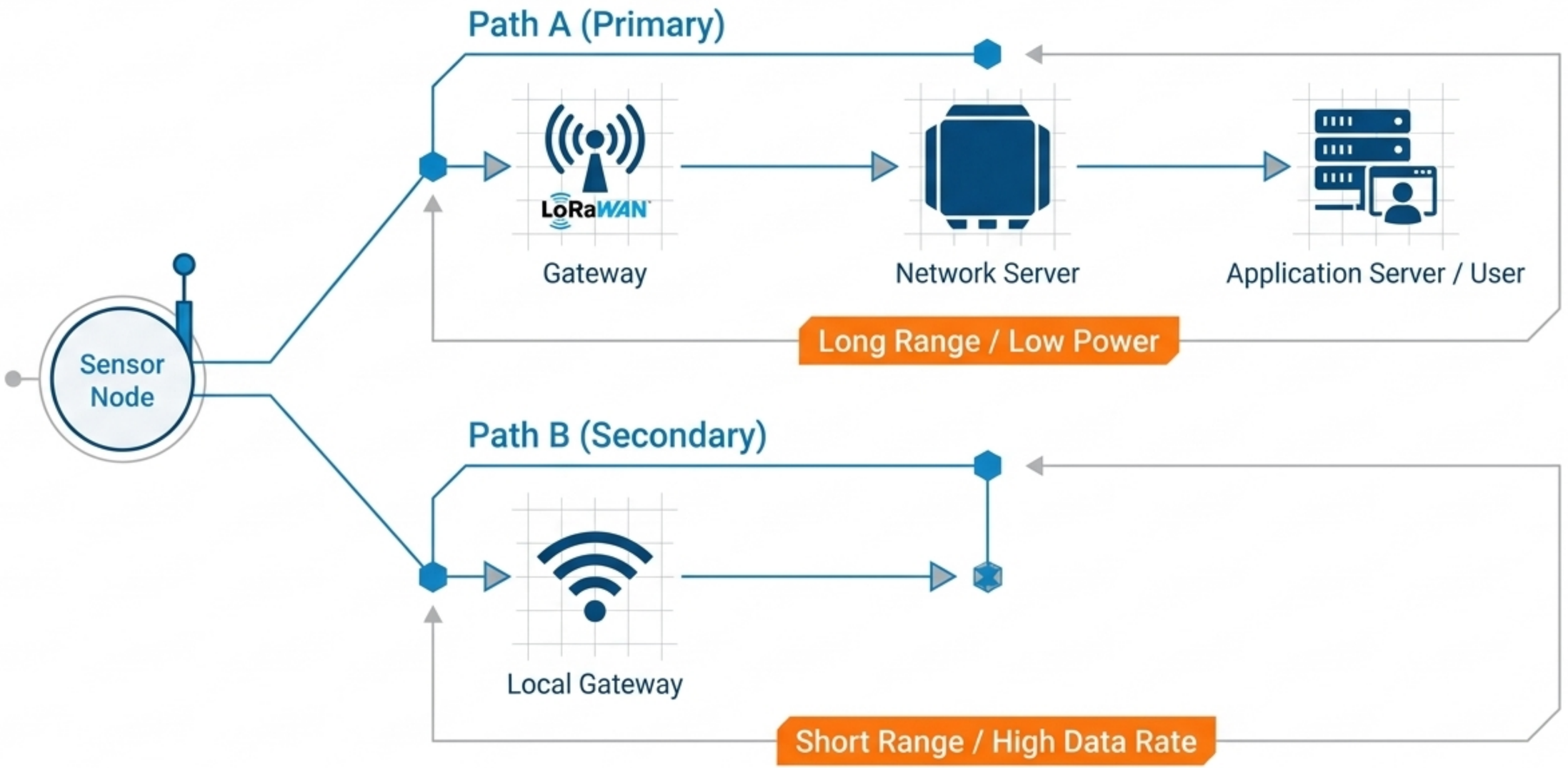


Temperature + RH
→ **Calibration Compensation**

Temperature and **Relative Humidity (RH)** act as **functional inputs** for data correction, not just passive observations.

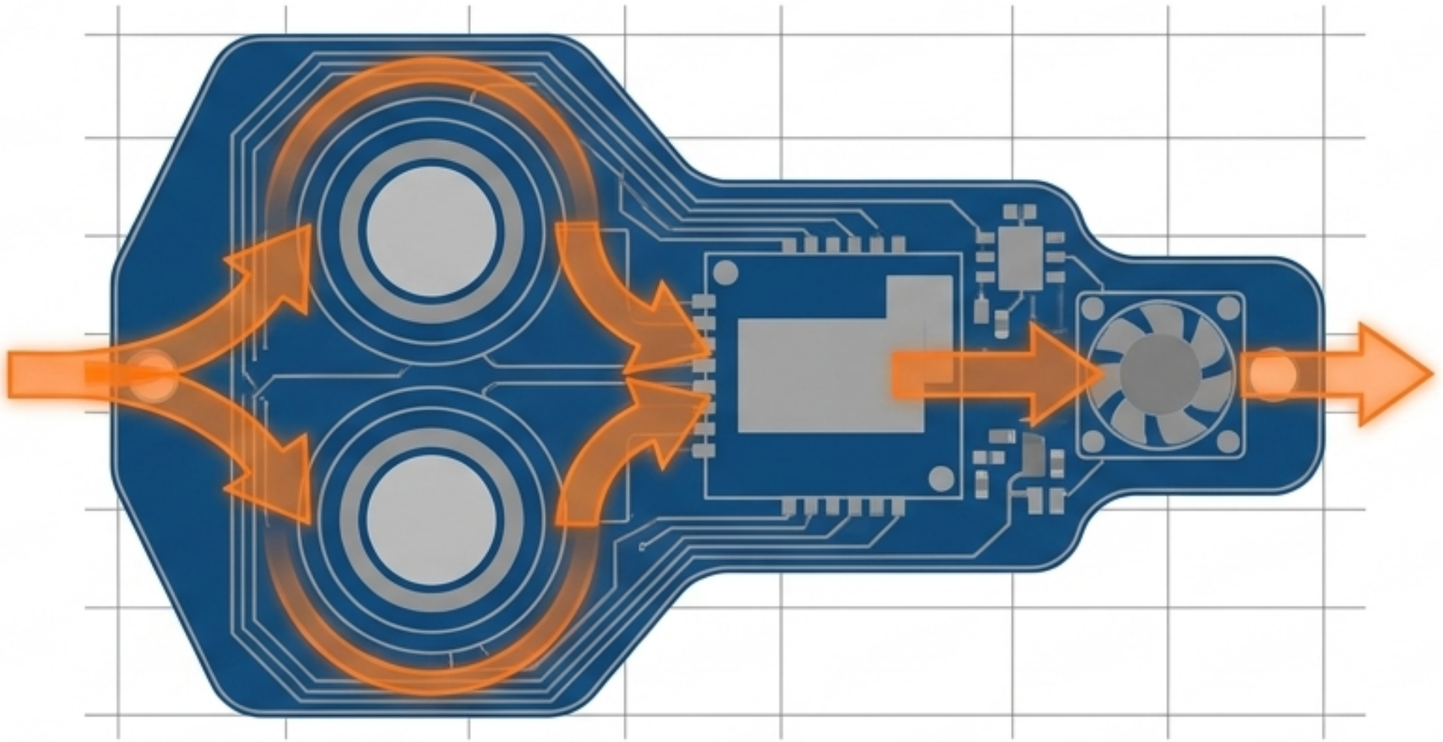
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Dual-Network Communication Topology



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Mechanical Design & Airflow Dynamics



9.2 cm^3
(Chamber Volume)

60 ms / reading
(Fan ON Duration)

Air In → CO / NO_2 sensors → PM sensor → Temperature/RH → Fan → Air Out

Source: Fig. 5 in the paper

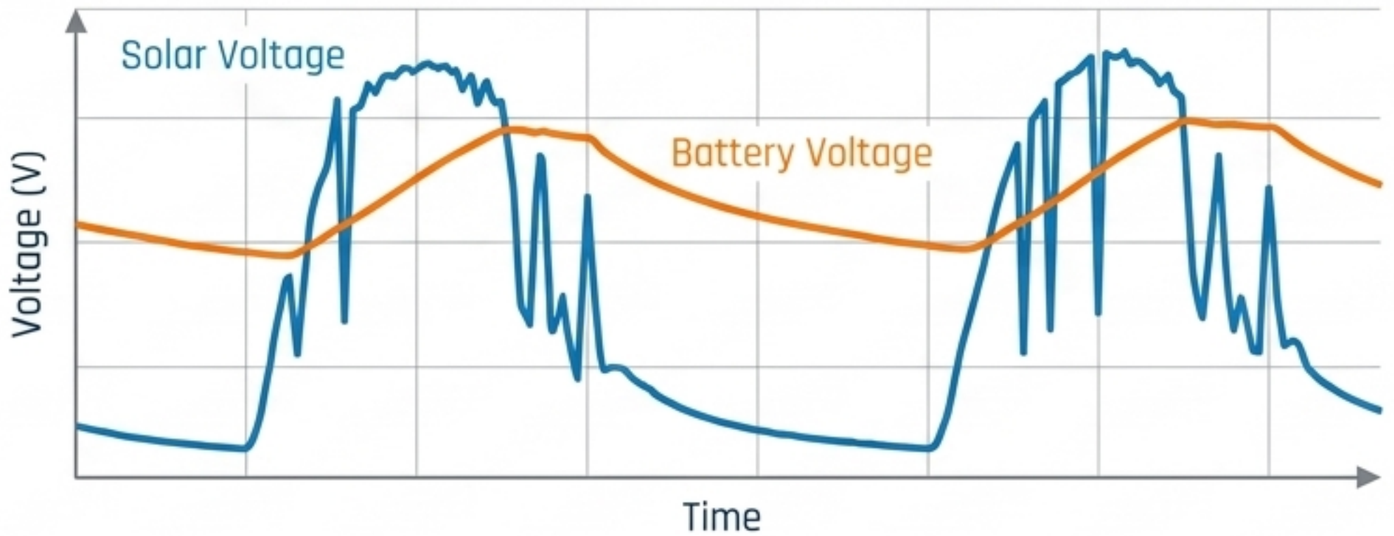
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Autonomous Power Management



~94 mAh/day

(Consumption)



<10%

(Max discharge depth)

Source: Fig. 7 in the paper.

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Field Calibration Deployment

Co-locating low-cost hardware with regulatory-grade infrastructure.



Test Subjects:
3 Sensor Nodes
(S1, S2, S3)

Ground Truth:
Teledyne API Model 300-A
Reference Station

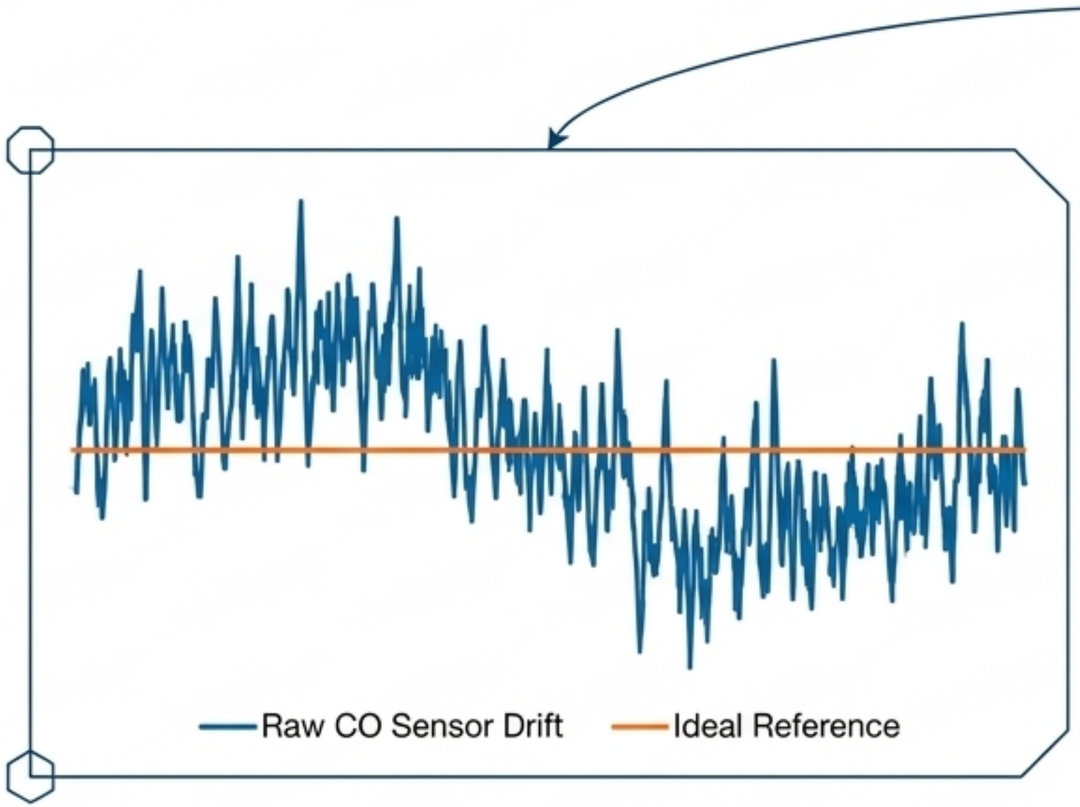
Sampling: **10-minute**
average readings

Location:
Khyber Pass, Auckland

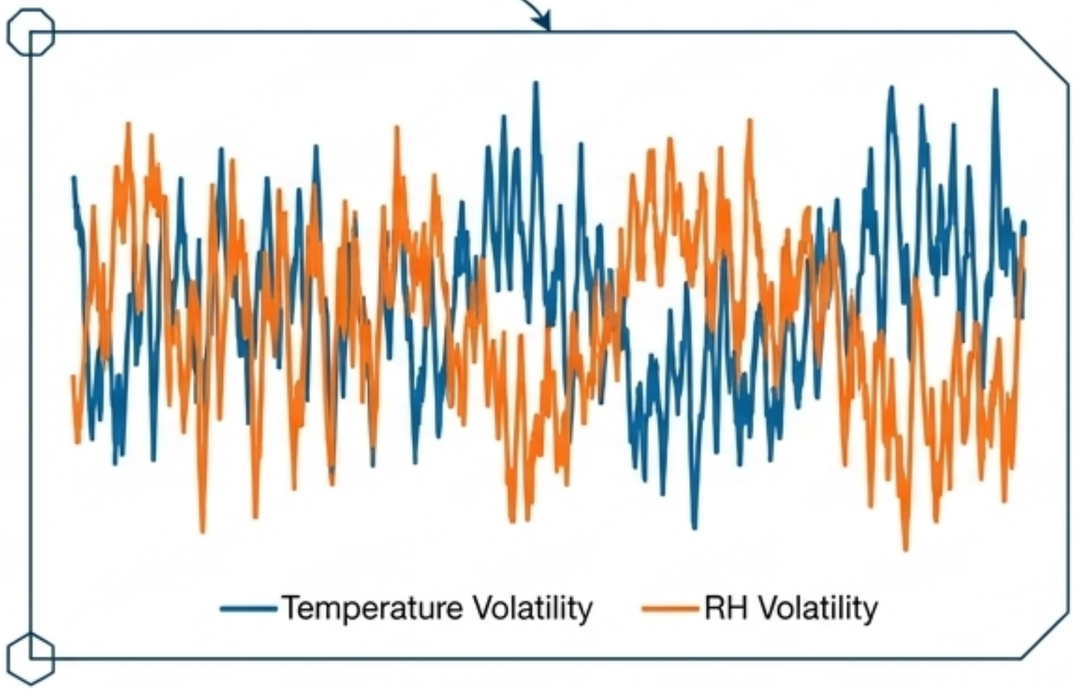
Source: Fig. 8 in the paper

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The Calibration Challenge



The Hardware Flaw:
Electrochemical sensors drift based on ambient environmental changes.

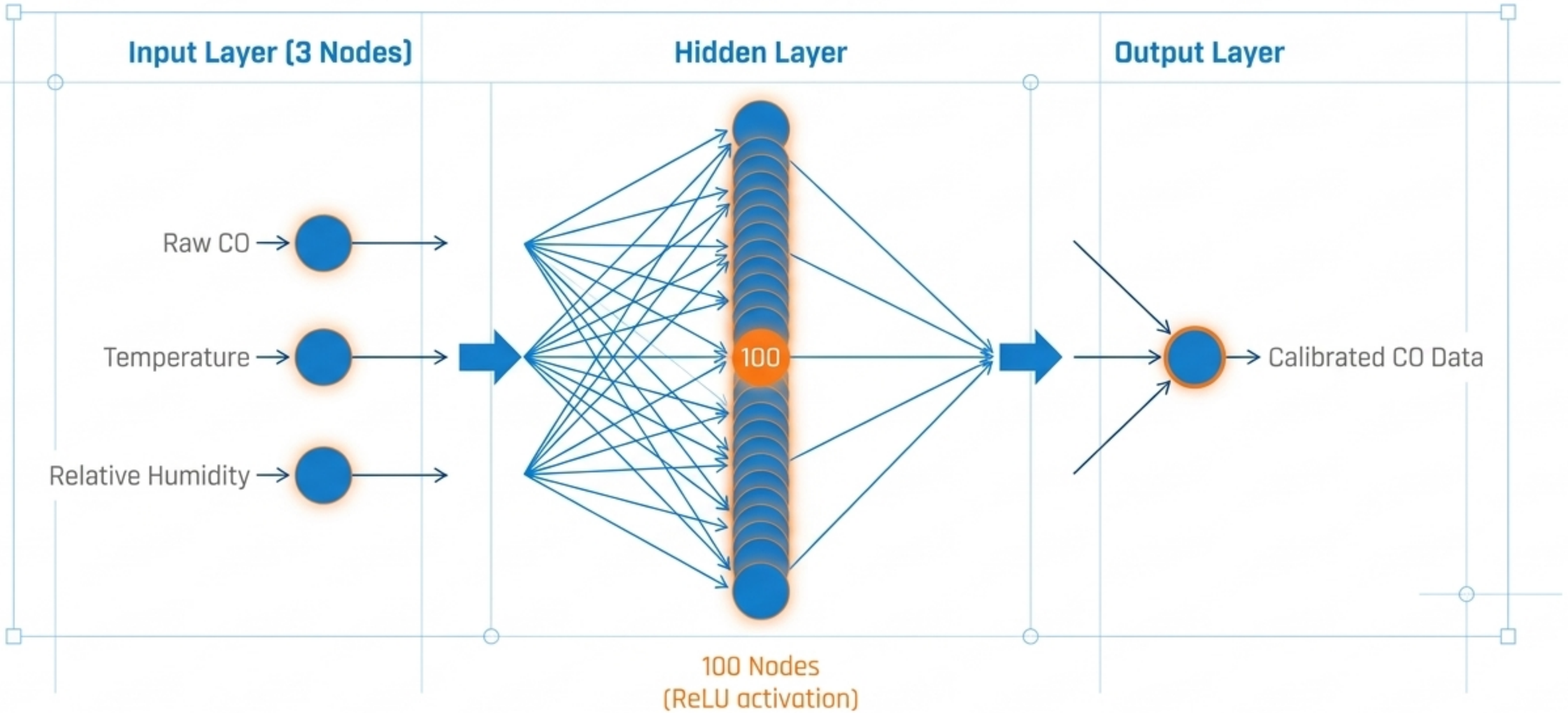


The Software Solution:
Requires Offset/Gain Calibration + Non-linear Temp/RH Compensation.

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Applied Machine Learning (ANN)

A shallow, single-hidden-layer Multilayer Perceptron mapping raw ambient data to reference truth.



Source: Fig. 12 in the paper

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The Climax: Calibration Results

OLS (Ordinary Least Squares)

Method: Linear mapping, no Temp/RH compensation.

Method: Linear mapping, no Temp/RH compensation.

MLR (Multiple Linear Regression)

Method: Linear, includes Temp/RH.

Results: MAPE: **48.71%**

R^2 = **0.607**

ANN (Artificial Neural Network)

Method: Non-linear mapping.

38.89% MAPE

R^2 = 0.780

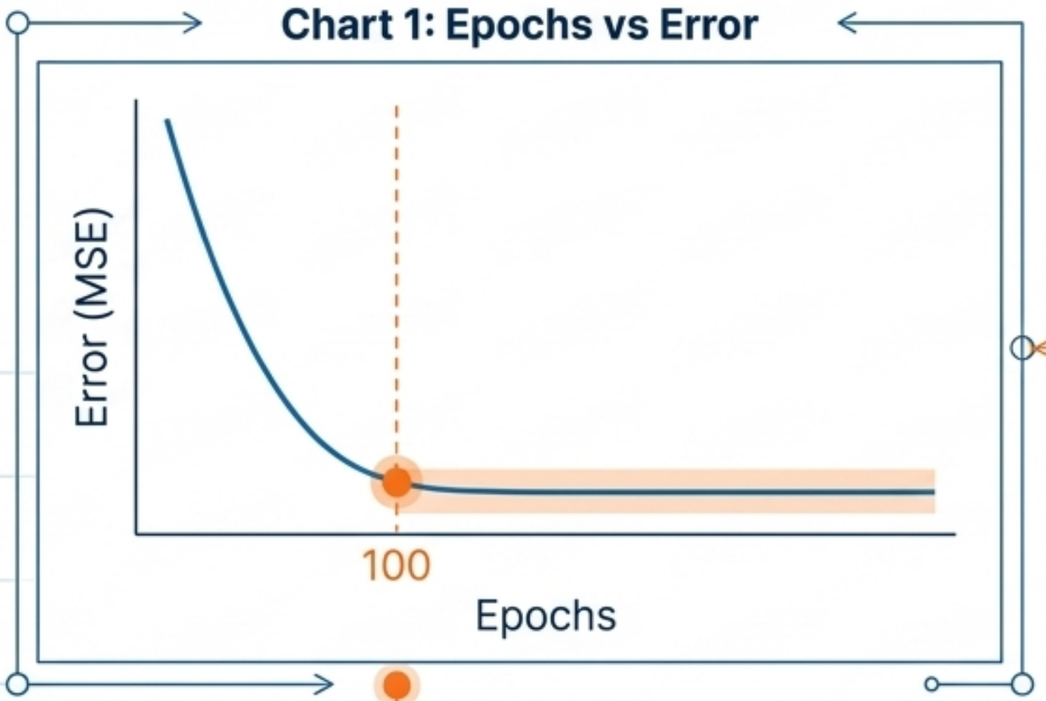
Source: Table III in the paper

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Model Optimization & Rigor

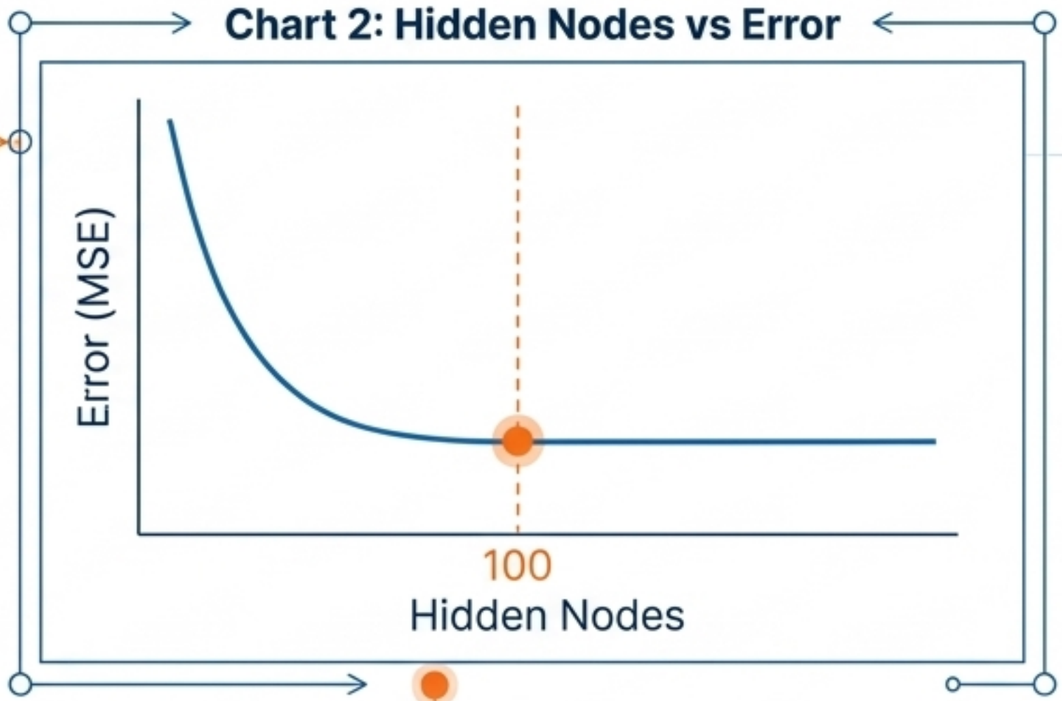
Hyperparameter tuning to maximize accuracy while preventing network overfitting.

Chart 1: Epochs vs Error



Diminishing returns after ~100 epochs.

Chart 2: Hidden Nodes vs Error



Optimal stability at 100 hidden nodes.

Source: Fig. 13 and Fig. 14 in the paper

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Conclusion & Scalability



Hardware

Validated an autonomous, ~$250 LoRaWAN Node.



Software

ANN significantly outperforms linear calibration methods.

Strengths & Limitations

✅ **STRENGTHS**

- ☑️ **Low Cost**: ≈ $250
- ☑️ **Integrated Design**: Sensor + IoT + Power
- ☑️ **Field Calibration**: Real-world co-location
- ☑️ **ANN Calibration**: Lower MAPE / Higher R^2
- ☑️ **Hybrid Communication**: LoRaWAN + Wi-Fi

⚠️ **LIMITATIONS**

- ⚠️ **Short Field Test**: 4–9 Sep 2019
- ⚠️ **CO Calibration Focus**: NO_2 / PM not yet calibrated
- ⚠️ **Long-Term Drift**: Not evaluated
- ⚠️ **Solar Validation**: Not extensively tested
- ⚠️ **Limited Dataset**: Generalization needs further testing

Source: Fig. 13 and Fig. 14 in the paper

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